ROUGH DRAFT

Beneath the Mesa

Assessing Aquifer and Well Water Resources Across Sandoval County, New Mexico

A County-Scale Screening Study with Cuba, New Mexico as a Focus Area

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ABSTRACT
Water scarcity is a persistent challenge across New Mexico. Sandoval County spans several hydrogeologic settings, and its communities depend on a mix of groundwater wells, surface water, and acequia irrigation systems. This project focuses on groundwater: it evaluates the condition of the county's aquifers and wells, with Cuba, New Mexico, on the eastern edge of the San Juan Basin, as a focused case study. The project is motivated by concern about declining groundwater near Cuba, but rather than assume how severe that decline is or what is causing it, we test what the available public data actually show, and where they are too sparse to say. *[Add the data used, the main findings, and what they mean, in three to four sentences, once the analysis is complete.]*

1 Introduction

1.1 Background and Motivation

[Background on water resources in New Mexico and Sandoval County, and the concern about declining groundwater near Cuba that motivated the project.]

1.2 Approach

The study works at two levels:

- **County-wide screening.** Map and compare aquifers, wells, and groundwater levels across Sandoval County. Because the county crosses several geologic regions, comparisons are grouped by aquifer unit instead of treating the county as one uniform aquifer.
- **Cuba case study.** A closer look at the Ojo Alamo Sandstone and San Jose Formation near Cuba, using published maps of both formations, aquifer-test records from nearby wells, and stream measurements on the Rio Puerco at Cuba.

1.3 Research Questions

RQ1. How have groundwater levels changed at monitored wells in Sandoval County, and which wells show sustained decline, stability, or recovery?

RQ2. What do well records show about well depth, depth to water, and the aquifer units that wells in different parts of the county draw from?

RQ3. What do available records show about groundwater near Cuba, and are they enough to evaluate the concern that its main aquifer is approaching severe depletion?

RQ4. How do the Ojo Alamo Sandstone and San Jose Formation compare near Cuba in extent, thickness, and hydraulic properties?

RQ5. Where is monitoring coverage strong, and where are the most important data gaps?

RQ6. Which indicators can be combined into a dashboard that shows groundwater stress without overstating how certain the data are?

The project proposal also asked about groundwater quality, acequias, and links between groundwater and surface water. Those questions were dropped because the data needed to answer them were not available (Section 7).

Figure 1 (to be added by the analysis lead): county map showing water-level monitoring wells, aquifer-test wells colored by formation, and the Rio Puerco gage, with Cuba marked.

Figure 1. *[Caption to be written.]*

2 Study Area

2.1 Sandoval County

[Describe the county, its hydrogeologic settings, and its main communities and water sources (Fig. 1).]

2.2 Cuba Focus Area

Cuba lies on the eastern edge of the San Juan Basin. The case study focuses on the Ojo Alamo Sandstone and San Jose Formation near Cuba (U.S. Geological Survey, 2020; Kernodle, 1996). *[Describe Cuba, its water supply, and the aquifers it draws from, and define the boundaries of the focus area.]*

3 Data

Town-specific data for Cuba are limited, so the study relies on broader public datasets that can be filtered geographically.

3.1 Datasets Used

USGS monitoring statistics. USGS monthly and yearly statistics for 30 monitoring sites in Sandoval County (17 stream gages, 9 wells, 2 canals, and 2 reservoirs), with records from 1926 to 2026. The file covers streamflow, depth to groundwater, reservoir storage, suspended sediment, water temperature, and water-quality measures: specific conductance, dissolved oxygen, pH, and turbidity. *[Source to be added.]*

OSE aquifer tests. New Mexico Office of the State Engineer aquifer-test records, 3,455 statewide and 179 in Sandoval County, including well depth, screened interval, geologic formation, and hydraulic properties such as transmissivity and storativity. *[Source to be added.]*

3.2 Datasets to Be Collected

- **Aquifer geometry near Cuba:** USGS San Juan Basin digital aquifer data for the Ojo Alamo Sandstone and San Jose Formation (U.S. Geological Survey, 2020).
- **More water-level records:** the National Ground-Water Monitoring Network (U.S. Geological Survey, 2026) and the Healy Collaborative Groundwater Monitoring Network (New Mexico Bureau of Geology and Mineral Resources, 2026).
- **County boundary:** a Sandoval County boundary layer to organize the maps.

[Data acquisition lead: explain why each dataset was chosen and which research questions it answers.]

4 Methods

4.1 Data Cleaning

Raw downloads are kept unedited in the repository. Each cleaned file is made by a script that reads the raw file and writes the cleaned one, so anyone can rerun it and get the same result. All cleaned files share the same columns for site ID, site name, latitude, longitude, date, and source, with one row per measurement and missing values left blank. *[Data cleaning lead: describe each cleaning step, the decisions made, and the final row counts.]*

4.2 Analysis

The analysis focuses on measurable indicators, such as depth to water, water-level change over time, well depth, aquifer unit, and hydraulic properties from aquifer tests, rather than a full hydrologic model. Where records are sparse, the results say so instead of filling the gaps. *[Analysis lead: describe the method used for each research question.]*

4.3 Classifying Findings

Each finding is labeled as one of the following:

- **Measured:** a trend or pattern the data show directly.
- **Hypothesis:** a possible explanation the data are consistent with but don't prove.
- **Insufficient data:** a question the available records can't answer yet.

4.4 Interactive Dashboard

Results are presented in an interactive Streamlit dashboard built with pandas and Plotly. Code and documen-

Figure 2 (to be added): time series of groundwater levels for wells with repeated measurements.

Figure 2. *[Caption to be written.]*

tation are maintained on GitHub (Djordjevic et al., 2026). *[Dashboard lead: describe each page of the app and what a visitor can do there.]*

5 Results

5.1 Groundwater Levels

[RQ1 (Fig. 2).]

5.2 Wells and Aquifer Units

[RQ2.]

5.3 Cuba Case Study

[RQ3 and RQ4. Include the Rio Puerco at Cuba stream measurements as surface-water context, not as evidence about groundwater.]

5.4 Monitoring Coverage and Data Gaps

[RQ5.]

6 Discussion

[What the results mean for Sandoval County and Cuba, and which indicators best communicate groundwater stress (RQ6).]

7 Limitations

The project proposal included three questions that this report does not answer, because the data needed for them were not obtained during the project:

- **Groundwater quality and salinity.** No groundwater-quality records were collected. The water-quality measurements in the USGS file all come from streams, mainly the Rio Grande below Cochiti Dam, far from Cuba.
- **Acequias.** No data on acequia locations or service areas were collected.
- **Groundwater and surface-water links.** Comparing the two requires repeated water-level measurements near a stream. The only stream records at Cuba are 10 visits to the Rio Puerco gage, and the nearest well in the water-level datasets is about 24 miles away.

[Add the other data gaps and what the data can't support.]

8 Conclusions

[Summary of findings and recommended next steps.]

Data and Code Availability

Code and data
github.com/pdjordj11/eng_220_aquifer_project
Interactive dashboard
eng220aquiferproject.streamlit.app
Documentation
pdjordj11.github.io/eng_220_aquifer_project

Author Contributions

P. Djordjevic: documentation, final report, presentation, and video.
D. Gonzales: data acquisition and dataset justification.
J. Wauneka: data cleaning and preprocessing.
M. Watkins: analysis code and visualizations.
M. Yousef: GitHub workflow and the Streamlit app.

Acknowledgments

[To be written.]

References

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